

LOGAN THOMAS

(510) 205-8343 | loganrichthomas@gmail.com | linkedin.com/in/loganthomas29 | loganrthomas.github.io | Alameda, CA

EDUCATION

UNIVERSITY OF CALIFORNIA, LOS ANGELES (UCLA) | Bachelor of Science, Mechanical Engineering | GPA: 4.0 Expected Jun 2029

SKILLS

Design & Modeling: SolidWorks (CSWA), AutoCAD, Design for Manufacturability (DFM), GD&T / Tolerance Analysis, Rapid Prototyping
Simulation & Analysis: FEA (SolidWorks Simulation), MATLAB, Simulink / Simscape (Model Calibration), Python (Data Analysis)
Data & Machine Learning: scikit-learn (Random Forests), Model Validation (Cross-Validation, ROC-AUC), Python Data Analysis
Manufacturing: Manual Machining, Waterjet, Laser Cutting, Sheet Metal Forming, Silicone Molding, Mechanical Assembly
Electronics & Testing: Circuit Design, Embedded Systems (Arduino), IMU Integration (MPU-6050), Instrumentation, Design of Experiments (DOE), Root Cause Analysis (RCA), Instrumented Thermal Testing, Technical Documentation, Soldering

EXPERIENCE

ENGINEER INTERN | DOER Marine | *Manned Research Submersible* Jun 2026 – Present

- Thermal Test Rig & Instrumentation:** Designed and built an instrumented test rig to benchmark two battery-wall cooling designs, iterating a foil-and-foam camera enclosure to remove thermal-reflection artifacts across 8 controlled heating trials.
- Thermal Model Calibration:** Performed initial calibration of a Simscape/MATLAB heat-transfer model against trial data, tuning the convection coefficient ($50 \rightarrow 70 \text{ W/m}^2\text{K}$) and adding a neoprene radiation layer to align simulated and measured cooling rates.
- Structural FEA & Load Path:** Ran SolidWorks FEA on a load-bearing deployment-rig frame, tracing a bending-driven stress concentration to a gap in the load path and closing it to cut peak von Mises stress from 33.3 to 1.27 ksi.
- Test Procedure & Verification:** Authored the formal thermal test procedure defining setup, active phases, and per-trial data logging for full repeatability, and verified component voltage and power specifications against the vehicle schematic.

ENGINEERING RESEARCH INTERN | ¹BL (UCLA Bioelectronics Lab) | *Ferrobatic Viral Testing Platform* Jan 2026 – Present

- Injection Port Integration:** Developed a mass-producible molding process embedding custom 2-mm self-sealing RTV silicone ports into testing cartridges, using root-cause analysis of mold geometry and fill sequencing to prevent recurring air-entrapment defects.
- Precision Enclosure:** Engineered a modular enclosure with a precision-guided sliding cartridge, achieving sub-millimeter vertical tolerances to preserve optical line-of-sight and unrestricted ferrobot kinematics.
- Optical Subsystem:** Re-architected the imaging assembly for interchangeable camera modules, designing a custom diffusion solution to eliminate hotspots and ensure uniform illumination across the microfluidic observation region.

Independent Research Project — Microfluidic Dispenser Characterization & ML Modeling

- End-to-End ML Pipeline:** Built an end-to-end characterization pipeline spanning experimental design, chip fabrication, 369-trial data collection, and MATLAB image analysis to map dispenser geometry to droplet output.
- Predictive Modeling & Validation:** Trained a Python random-forest model predicting droplet volume to $0.369 \mu\text{L}$ cross-validated error, diagnosing overfitting via the CV/train ratio and building an inverse-design tool for geometry selection.

TECHNICAL PROJECTS

SELF-DEPLOYING LAPTOP STAND | Personal Project | *Mechanism Design, Statics & FEA* Jun 2026 – Present

- First-Principles Statics:** Derived the linkage statics symbolically and proved by contradiction that no equilibrium exists under load, identifying the hard stop as the primary load-bearing member.
- FEA Validation & Iteration:** Modeled the assembly in SolidWorks Simulation with pin connectors and contact at the load path, using a mesh-convergence check to confirm a real stress concentration at the stop, and separately hollowed and ribbed the platform, cutting mass 46% while recovering $3\times$ the stiffness lost to hollowing.

ELECTRIC GUITAR FABRICATION | Personal Project Jun 2024 – Present

- Precision Manufacturing:** Fabricated two custom electric guitars using manual machining (router/bandsaw), achieving precise scale length and fret alignment to guarantee intonation accuracy across the fretboard.
- System Integration:** Installed and calibrated a Floyd Rose tremolo bridge, balancing spring tension against string gauge to ensure precise pitch return after mechanical modulation.